[ ]:
```
#for loops
```

[1]:
```
n = int(input("Enter a number: "))
e = 0
o = 0

for i in range(1, n + 1):
    if i % 2 == 0:
        e += ia
    else:
        o += i

print("Sum of even numbers:", e)
print("Sum of odd numbers:", o)
```

```
Enter a number:  3
Sum of even numbers: 2
Sum of odd numbers: 4
```

[4]:
```
'''
Count Digits in a Number Get an integer from the user and count the number of digits using a single for loop.
Do not convert the number into a string. Example: 58392 → 5 digits
'''

num = int(input("Enter a number"))
count = 0
if num == 0:
    count=1
else:
    for i in range(num):
        count+=1
        num = num//10
        if num == 0:
            break
print("number of digits ",count)
```

```
Enter a number 1234
number of digits  4
```

[12]:
```
'''
Find the Second-Largest Number Given a list of distinct numbers,
find the second-largest number without using sort() or max().
'''

li = [1,2,4,6,8,9]
max1 = li[0]
max2 = li[0]
for i in li:
    if (i > max1):
        max2 = max1
        max1 = i
    elif (i > max2):
        max2 = i
print("second maximum :",max2)
```

```
second maximum : 8
```

[16]:
```
'''
Count Vowels and Consonants Get a string from the user and count the vowels and consonants separately.
Ignore spaces, digits, and special characters
'''

text  = input("Enter a string")
v=0
c=0
for char in text:
    if char.isalpha():
        if char in "aeiouAEIOU":
            v+=1
        else:
            c+=1

print("vowels : ",v)
print("consonants : ",c)
```

```
Enter a string taehyung
vowels :  3
consonants :  5
```

[18]:
```
'''
5. Calculate Factorial Get a positive integer from the user and
calculate its factorial using a single for loop. Example: 5 → 120

'''
num = int(input("Enter a number"))
fact = 1
for i in range(1,num+1,1):
    fact = fact*i
print("factorial : ",fact)
```

```
Enter a number 5
factorial :  120
```

[27]:
```
'''
Find Numbers Divisible by 7 but Not by 5 Print all numbers between 1 and 200
that are divisible by 7 but not divisible by 5.
Also, count the matching numbers.
'''


count = 0
for i in range(1,201):
    if(i%7 == 0 and i%5!= 0):
        print(i,end=" ")
        count+=1
print()
print("count:",count)
```

```
7 14 21 28 42 49 56 63 77 84 91 98 112 119 126 133 147 154 161 168 182 189 196
count: 23
```

[30]:
```
'''7. Calculate Electricity Bill Get the units consumed by each of five customers using a list.
Calculate each customer's bill using these rates:

* First 100 units: ₹2 per unit

* Next 100 units: ₹3 per unit

* Above 200 units: ₹5 per unit
'''

units = [80,100,150,230,300]
for i in range(5):
    bill = 0

    if units[i] <100:
        bill = units[i]*2
    elif units[i] <200:
        bill = 100*2 + (units[i]-100)*3
    else:
         bill = 100*2 + 100*3 + (units[i]-200)*5

    print("customer",i+1)
    print("units ",units[i])
    print("bill ",bill)
    print()
```

```
customer 1
units  80
bill  160

customer 2
units  100
bill  200

customer 3
units  150
bill  350

customer 4
units  230
bill  590

customer 5
units  300
bill  800
```

[34]:
```
'''
Reverse a Number Get an integer from the user and reverse its digits using a single for loop.
Do not convert it into a string. Example: 12345 → 54321
'''
num = int(input("Enter a number"))
reverse = 0
for i in range(num):
    digit = num%10
    reverse = (reverse*10)+digit
    num//=10
    if num == 0:
        break
print("reverse number ",reverse)
```

```
Enter a number 123
reverse number  321
```

[37]:
```
'''
9. Analyze Student Marks Given a list of marks, count the students who passed (50 or above),
failed (below 50), and scored distinction (75 or above).
'''
marks = [90,45,67,78,40,98]
p,f,d = 0,0,0
for m in marks:
    if(m >= 75):
        d+=1
    elif m>=50:
        p+=1
    else:
        f+=1
print("total distinction students ",d)
print("total passes students ",p)
print("total failed students ",f)
```

```
total distinction students  3
total passes students  1
total failed students  2
```

[42]:
```
'''
Find Perfect Numbers Print all perfect numbers between 1 and 1000 using a single for loop.
A perfect number equals the sum of its proper divisors. Example: 6 is perfect because 1 + 2 + 3 = 6.
'''
for  num in range(1,1001):
    sum = 0
    for i in range(1,num):
        if(num % i==0):
            sum+=i
    if(sum == num):
       print(num)
```

```
6
28
496
```